



Aston University

Integrated Healthcare Analytics: A Unified Dashboard and Conversational AI Platform for Hospital Management

A dissertation submitted in fulfilment of the requirements for the degree
of
Master of Science in Data Science.

September 2025

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Word count: 9634 (excluding references)

Acknowledgment

I wish to extend my deepest gratitude to all who have supported me throughout the development of this project.

I am especially indebted to **Jeremy Levesley**, my supervisor, for his exceptional guidance, encouragement, and constructive feedback, which have been central to shaping the direction and quality of this research.

My sincere thanks also go to **Matthew Stuart Bickley**, my programme director, and to the faculty and staff of Aston University for providing the resources, facilities, and an enriching academic environment that made this work possible.

I also extend heartfelt appreciation to my family and friends for their patience, understanding, and unwavering support, which gave me the perseverance to complete this study.

Finally, I acknowledge the invaluable contribution of publicly available datasets and research papers, which served as a foundation for this work and exemplify the collaborative nature of scientific research.

This project has been both challenging and rewarding, and I remain sincerely thankful to all who contributed to its successful completion.

Sathish Kumar Kaliappan

Abstract

Across the globe, healthcare systems are experiencing extreme pressure to promote patient outcomes while working with budget constraints, rising need, and an ageing population. Traditional models for the monitoring of hospital performance – normally static reports or standalone dashboards, provide limited view of the complex operational reality. This dissertation presents the design, implementation and evaluation of a unified healthcare analytics platform that offers a hospital data dashboard alongside a conversational AI chatbot assistant. The motivation for this study was to address the areas of concern described above by integrating finance, patient outcomes and workforce metrics on one platform to increase decision support, situational awareness and interpretability for hospital administrators and clinical managers.

A multi-tab dashboard was created to visualize operational and clinical data including budget versus actual spending, procedure cost distributions, patient flow funnels, readmission risk bubble plots, and staff allocation metrics. In addition, a natural language chatbot using a large language model was also incorporated to provide users with conversational access to the query database, they are then provided with explanatory narratives of the KPI's (key performance indicators), and where relevant, deidentified patient-level data. Due to not having ethically accessible clinical data, a combined dataset was created with publicly available hospital data, and synthetically generated data that matched substantially similar distributions. Development used Python, Streamlit, Plotly and NLP API while adhering to recommended practices in user-centered dashboard design, privacy-preserving data engineering and conversational analytics.

To evaluate the system we compared dashboard-only interactions with chatbot-assisted-based use cases to determine if the levels of interpretability, usability, and decision support was enhanced. Furthermore, I contextualized the work as recent literature of hospital dashboards, synthetic health data, and text-to-SQL conversational systems. As such, this dissertation makes an original contribution to the expanding field of dual-interface healthcare analytics. The findings indicate that visual dashboards integrated with conversational AI supports a reduced time-to-insight and subsequently, more users can interrogate complex data sets without domain expertise. The work demands future directions of predictive analytics, clinical ontology integration, and systematic user evaluation further validate the ability of the system to scale and be effective in real-world hospital systems.

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Chapter 1

Introduction

1.1 Context & Rationale

Health systems across the world are experiencing increasing pressure to provide safe, efficient, and equitable care while utilizing limited resources. Tremendous demographic shifts, with rapidly increasing ageing populations and rising rates of chronic conditions, have both added hospital volumes and increased the complexity of the pathways of care [27, 34]. In England alone, 2023 elective waiting lists were over 7.6 million patients; the growth rate of healthcare spending cannot keep pace with population needs [32]. Likewise, similar trends have been documented globally, which highlights a common need for data informed resource allocation and operational visibility [26].

In this context, hospitals are starting to implement digital health tools to improve monitoring ability and decision-making pathways. Of these opportunities, interactive dashboards and conversational AI assistants show potential as complementary innovations. Dashboards can be used for high-level, at-a-glance monitoring of metrics such as budgets, patient throughput and staffing ratios [19, 11]. Conversely, AI-informed chatbots can act as a natural-language interface to complex datasets reducing the barriers of accessing more detailed information [37, 6]. Together, they should support a shift from report based, to continuous, end-user-centric analytics - in the management of health-care.

1.2 Challenges with Traditional Hospital Monitoring Practices

Traditional hospital reporting practices relied on periodic, static summaries, the summaries are often fed off of excel, PDF, or other reporting exported from business intelligence teams. These methods have long lag times, limited interactivity, and steep learning curves for non-technical personnel. Standalone dashboards have improved those aforementioned drawbacks, but still typically provide descriptive, not explanatory or predictive information [7]. Users can see the KPIs, but they do not know "why" or how it was developed, nor do they have easy access to the data patient or unit-level breakdowns.

Another barrier is user access to data. Clinical and operational datasets are usually siloed, or under austere privacy protection regimes, and it is hard to conduct compre-

hensive analyses. Clinical metrics and staffing levels are often omitted from publicly available datasets, and comprehensive insight into pressures across finance, patient outcomes and workforce is extremely difficult to obtain [15].

1.3 Rationale for Integrated Dashboard – Chatbot Systems

The bigger purpose of this project is to test the hypothesis that a hospital performance dashboard and conversational AI chatbot, working together, can overcome these contemporary barriers. The dashboard would give the user visualisation and the ability to rapidly scan KPIs across budgets, patient outcomes and staffing. The chatbot component would enable the user to ask follow-up questions in plain English and receive either SQL-driven responses or narrative descriptions without needing to know SQL and without making it necessary to provide hard to understand graphical outputs.

In integrating the two components, the system has the potential to:

1. Promote situational awareness by integrating multi-domain hospital KPIs [9];
2. Improve interpretability by enabling to formulate follow-up questions on why and how KPIs were established [1];
3. Inform decision making by expediting actionable metrics as well as low tech access to financial and staffing data [16].

1.4 The Use of Synthetic Data

As is well known in the hospital analytics domain, the ethical and legal barriers to sharing and utilizing patient privacy data remains the main barrier to hospital analytics advancement. Synthetic data provides a practical alternative. Creating realistic, not identifiable records enables a researcher to utilize their analyses platform to create and train their AI models and show proof of concept without compromising confidentiality [29, 8]. This project combined an non-identifiable synthetic data set with publicly available hospital datasets (e.g., Kaggle HK Vitals) to create a comprehensive dataset with hospital admissions, procedures, outcomes and staffing levels, which is similar to a growing trend in digital health for privacy-by-design analytics [14].

1.5 Aims and Objectives

The purpose of this dissertation was to develop, create and evaluate an integrated analytics solution that combines a hospital performance dashboard with an AI conversational chatbot assistant. This can be delineated into five specific aims.

Data – Integration: Aggregate anonymised synthetic hospital data–finance, patient outcomes and staffing–into a single analytic file.
Dashboard Development: Create an in-

teractive dashboard that displays KPIs, for example, budget to actual spending, patient flow, readmissions, and staffing models. Conversational – AI integration: Integrate a natural language chatbot that allows users to ask data queries and receive direct responses to their queries and visualisations about KPIs. Evaluating: Evaluate the combined system with standalone dashboard use in member of senior executive leadership, etc. measure usability, interpretability and decision support functions [4, 13, 12].

1.6 Research Questions

The study inquiries into three blended research questions:

1. Visualisation Effect: How might a consolidated analytics dashboard increase visibility of hospital KPIs pertaining to finances, patient outcomes and staffing?
2. Conversational Complement: How might a conversational bot aide visualisation within the dashboard by providing interpretive explanation of data and access to data for non-coding users?
3. Comparative Usability: How does the combined dashboard-bot approach compare to currently available data reporting systems with respect to usability and proposed decision support of hospital management?

1.7 Knowledge Contribution

The contributions of this project to the domain of digital health analytics are discussed in specific ways:

Integrated Platform: It demonstrates a dual-interface visualisation that capped with conversation exploration in a unified interface [36, 21]. User-Centred Design: It applies modern human-computer interaction principles - colour-coded KPI cards, organized structure, and consistent categorization and labeling [23, 22]. Operational Prompt Engineering: It develops an operational prompt template to convert plain English questions into SQL queries with date and budget conditions [35, 39].

Chapter 2

Literature Review

2.1 Introduction to the Landscape of Literature

Literature about hospital analytics and digital decision support exploded after 2018, as the world transitioned to more data-driven healthcare solutions [2]. Researchers developed dashboards in the context of performance monitoring, conversational interfaces to access data, and the development of synthetic datasets to allow privacy-secured analytic. Yet, not much literature has combined all three threads in a single operational setting. This chapter will assess these three areas, the evolution of these areas, and the knowledge deficits in each area that this study addresses.

2.2 Evolution of Digital Dashboards in Hospital Management

The use of dashboards to assist hospital management began in the early 2000s, but relatively rapidly moving from static dashboards to interactive, real-time analytics [17]. Present day dashboards include a variety of data types - finances, admissions, discharges, readmissions, staffing, and patient outcomes - which are displayed in visually accessible formats such as funnel diagrams, waterfall charts and bubble plots [19].

Interactive dashboards have been claimed to improve situational awareness, lessen information fatigue and allow anomalies to be detected sooner [9]. A simple waterfall chart can show the budget variance from contract to contract fairly quickly, funnel chart can present where a bottleneck exists relative to patient flow, and bubble plots can demonstrate the risk-adjusted rates of readmission patient to patient and across multiple conditions [7]. This visualisation has a comparable trajectory to the emergence of health informatics "cognitive dashboards," the example must use action outputs instead of just representing measurements [11].

Though useful when relevant, dashboards limit in that they are only descriptive measures. [7] further highlight that whether the visualisation floods metrics that change, they can still display them but do not provide the context how the metrics changes were connected to the casual theory. For instance, some analytic expertise is still warranted from the user to a complex visualisation, and/or to filter through some measures [15]. Hence, researchers have supported this hybrid dashboard that coexist with functional interface overlays [21].

2.3 Conversational Interfaces, and LLM's for Health Analytics

Conversational interfaces, most specifically LLM's, provide an alternative approach for clinical staff to work with data. Instead of employing a fixed parameter set for filters, the clinician or administrative user can simply ask a question in normal everyday language request a structured output from the dashboard [37]. LLM's lower the technical barrier, especially for users with minimal or no analytic backgrounds to conduct ad hoc inquiries minus expertise [6].

There is evidence that dashboards with chatbots improve user satisfaction. [21] illustrated how if users could 'speech' to the dashboard interface, participants completed their analytics task in record time, and with less errors, compared to only using dashboards. Users engage metrics as 'why' conversation engage that KPI, bring engage from an elaboration that metric even measures is not present. However, if the user needs to coordinate with another specialty it impedes with a agent [31].

There are still important challenges. The LLMs can ramble, miscontrue the meta-date/schema mapping to create invalid SQL code [35]. There also privacy concern about audits and assigning right roles in the LLM and hospital DB [14]. Thus, we recommend the use of prompt templates, validated queries, and reading only in the Health Systems.

2.4 Synthetic Health Data and Privacy Preserving Analytics

One challenge in health informatics include all the limitations of access to patient-level data and often more on privacy. Synthetic Health data - or, algorithms to imitate genuine patient data distribution- is a fascinating avenue to began addressing problems. In using synthetic approach range from basic parametric samples, to complex generative adversarial networks (GANs) and VAEs that develop trajectory models of patients multidimensional state [29, 24].

[8] proposes 3 potential benefits of synthetic data advanced health analytics:

- provide privacy when there are no patient identifiers;
- provide scale by data space larger than just one organisation, and
- reproducible by open development without sacrificing confidentiality.

Synthetic data also presents its ongoing concerns and limitations. In an effective synthesis, and in general if the base data used to generate the model was biased and its want be imperfect then the synthetic data could continue the bias [25]. Also, rare by clinically significant outputs can be subject to bias. Models fidelity can deteriorate from

exclusively on synthetic data, Thus there continued development to link the synthetic and anonymized public data, where test to rest in alignment with distributions [8].

2.5 Merging Dashboards, AI Conversations and Synthetic Data

The literature has arrived at the same insight: there can be no tool whose sole purpose is to meet hospital decision-maker needs. Dashboards offered visibility but have no contextual narrative; chatbots suggest narrative but are dependent upon a structured data backbone; while synthetic data is sufficient for development, it has to be validated appropriately. Very few studies are able to showcase all three pillars in one operational system.

Recent studies by [36] proposed a model for "dual-interface analytics", specifically leveraging cognitive dashboards along with natural language query engines. [37] piloted a text-to-SQL chatbot for hospital metrics but did not pair the chatbot with a dashboard. [29] developed a synthetic data pipeline for clinical trials but did not organize that with real-time analytics. This research fills these gaps - unifying dashboard visualisation, conversational AI and synthetic data generation in a singular study, which operationalizes the integrated perspective outlined in the literature.

2.6 Major Themes Evident from the Literature

Theme 1 - Descriptive to Explanatory

In the past five years, the transformation of simple descriptive dashboards to systems that describe explanatory narratives and predictive what if forecasts has evolved. Hospitals now want not only "what happened" but "why" and "what next" [19, 16].

Theme 2 – Natural Language as a Universal Interface

Recognition has been made that natural language interfaces can provide democratized access to data analytics. This trend follows general trends in AI where voice and text interfaces have displaced more rigid forms-based access systems [21, 33].

Theme 3 – Privacy-by-Design in Health Analytics

There has been increased interest in synthetic data and privacy-enhancing technologies due to increased pressure due to GDPR and HIPAA. Organisations are looking at synthetic data pipelines as requisite for prototyping and testing analytics tools [8, 14].

Theme 4 – Usability and Human Factors

Research in human factors show that mixed-mode systems (visual + conversational) reduce cognitive workload, and improve decision making under time pressure [36, 9]. The Technology Acceptance Model and ISO 9241 usability guidelines are being used to evaluate these tools [4, 13].

2.7 Research Gap and Justification for Present Study

There is a robust literature on dashboards, chatbots and synthetic data but integrated systems are rare. Most hospital dashboards are retrospective with an emphasis on compliance and reporting, as opposed to decision support [7]. Chatbots are developed as pilot systems without live data and visual KPI [37]. Synthetic data is not being used outside of research environments; however, it can be useful for privacy preserving analytics at scale [24].

This dissertation addresses that gap by:

Building an operational dashboard that includes multi-domain KPIs (finance, patient outcomes, staffing) [11]. Integrating a conversational AI chatbot that translates natural language into SQL queries against the same database [35]. Using synthetic data to ensure privacy while testing realistic hospital scenarios [8].

2.8 Conceptual Framework Guiding Study

Based on this literature, this study uses a conceptual framework whereby:

- Data Infrastructure is the platform for both dashboard and chatbot;
- Visual Analytics provide a means for rapid scanning KPI;
- Conversational Interface is the mode for explanatory and patient-level drill down;
- Synthetic Data preserves privacy and scale.

These integrated approaches to enable the operationalization of health informatics recent calls for multi-modal, user-centred decision support tools that are instrumental to ethical regulatory guidelines [27].

Chapter 3

Data

3.1 Overview of the Data Strategy

An effective data strategy is paramount to any hospital analytics initiative. This study purposely integrated publicly available hospital data with synthetic data to produce adequate coverage of finance, patient outcomes and staffing measures, while adhering to privacy regulations. This chapter outlines the sources, integration technique, data cleaning, and feature engineering that resulted in the unified dataset behind the dashboard and chatbot.

3.2 Data Sources and Justification

The base of the dataset utilized comes from the Kaggle HK Vitals repository, which contains anonymised records of hospital admissions, interventions, and outcomes. This source was chosen because it was publicly accessible, in a structured format, and had already been used for benchmarking hospital analytics systems. While, the public dataset lacked sensitive but important elements—such as cost structures and staff ratios—synthetic data was created to add the base data. Using both data sources aligns with best practice guidelines concerning privacy-preserving analytics, which recommended combining open data and synthetic data as a way to maintain the real-world complexity [8, 24].

3.3 Synthetic Data Generation

The synthetic data pipeline was built to provide plausible but non-identifiable records for three domains—finance, patient outcomes and workforce data. A blended approach was employed:

- **Parametric Modelling:** Costs per intervention and staffing amounts were derived from published national averages, using normal or log-normal distributions.
- **Rule-Based Logic:** For example, oncology and cardiology interventions were given higher weighted costs and longer mean lengths of stay, consistent with known care patterns in hospitals.
- **Derived Indices:** New metrics were developed (e.g. "Staff Shortage Index," "Waiting List Days," and "Ageing Population Share") from the synthetic data

base to support advanced KPIs.

Thus, while no real patient records were used, the distributions did approximate plausible hospital environments. The process was using privacy-by-design principles and random (non-sensitive) seed management for reproducibility [29].

3.4 Data Wrangling and Standardisation

Before loading into the analytics platform, all records—both public and synthetic—underwent thorough cleaning and standardisation:

Timestamps: Admission and discharge dates were converted to ISO 8601 format and validated for logical consistency (i.e. no stays less than 0, no dates in future). **Coding Harmonisation:** Procedure codes were aligned to a single coding schema to allow for cross-tabulation. **Currency and Units:** All financial fields were converted to GBP, systematically in decimal precision. **Missing Values:** When data was not available (e.g. patient age or staff amount) it was replaced with either mean substitution or rule-based substitution based on the variable.

By undertaking these steps, the project produced a clean, comparable dataset for time series analysis, cohort segmentation, and visualisation.

3.5 Feature Engineering and Key Indicators

For the dashboard and Chatbot, multiple derived variables were created:

- **Budget vs Actual Variance:** per procedure and aggregated annually to form waterfall charts
- **Patient Flow Indicators:** admission-discharge lag times, readmission flags or adjuvant comparisons and waiting list outcomes to form funnel diagrams
- **Readmission Risk Bubble Plot:** readmission rate, volume of cases, and case age to provide a three-dimensional risk depiction
- **Staff Tasking Metrics:** staff-to-patient ratios and staff shortages by condition to provide bubble/scatter plots of doctors' performance

These features contributed to the aim of interlinking finance, patient outcomes and workforce datasets in one combined analytic process.

3.6 Data Quality and Consistency Checks

The HK Vitals data from Kaggle were the base data; it had few rows and missing attributes. Therefore, synthetic data were created by generating new columns and expanding rows by logical rules. Values remained in a realistic clinical value range, IDs were ensured not to repeat, and any attribute relationship was honored. This fact kept the extended dataset consistent and reliable in the support of the dashboard and chatbot.

3.7 Ethical and Privacy Considerations

Synthetic data has few, if any, patient-sensitive details making it suitable to privacy regulations like GDPR and HIPAA. All records were either anonymised or fictional. The pipeline kept no ability to reverse back to a real patient. This is congruent with the increasing consensus that we can use synthetic datasets as a realistic and safe vehicle for innovation in hospital analytics without compromising patient privacy [29, 24, 14].

3.8 Limitations of the Dataset

Although the dataset was curated carefully, there are still limitations:

- **Synthetic:** These generated records may not capture all of the relationships and outliers that appear in the real world [25].
- **Temporal:** some fields were obtained cross-sectionally, preventing longitudinal tracking other than at an aggregated monthly level
- **Generalisability:** Patterns identified may not translate to hospitals in other countries

These limitations are conceded and will provide context for evaluations presented in coming chapters.

Chapter 4

Methods and System Design

4.1 Overview of the System Development Approach

This project was architected accounting for two complementary components in an integrated healthcare analytics platform: (1) An interactive Dashboard for visualising hospital key performance indicator (KPI); and (2) a conversational AI chatbot assistant to natural-language querying of the same data source. The method was grounded in user-centred design, modular software architecture and privacy-by-design data practices to align with current best practice in digital health innovation [23, 8].

4.2 Design Principles

All system design was guided by four principles:

1. **Modularity** – Data, analytics, visualisation and conversational components had to be logically distinct to promote maintainability and scalability.
2. **User-Centred** – User interface elements matched the mental models and workflow of hospital managers and clinicians [23].
3. **Transparency** – The tooltips, legends and explanatory text makes the KPI interpretable and trustworthy [31].
4. **Privacy-by-Design** – Synthetic data and secure storage were used to mitigate access to sensitive patient information [8].

4.3 Data Processing Pipeline

A reproducible pipeline written in Python (version 3.10.2) was coded to process both public and synthetic datasets into a usable format for the dashboard and chatbot. Key steps included:

- **Ingestion:** Loading each CSV file (the public datasets and synthetic enrichment) into a Pandas data frame.
- **Cleaning and Harmonisation:** Timestamp parsing, code normalisation, missing value imputation, currency conversions.
- **Feature Engineering:** Derived variables such as Budget Variance, Staff Shortage Index, Waiting List Days.

- **Data Loading (Chatbot):** For the chatbot, the dataset was converted into an SQLite database (.db), generated through Python, to be saved locally and stored for querying purposes.
- **Incremental Updates:** Scripted functions enabled redistributing new data for ingestion and processing purposes.

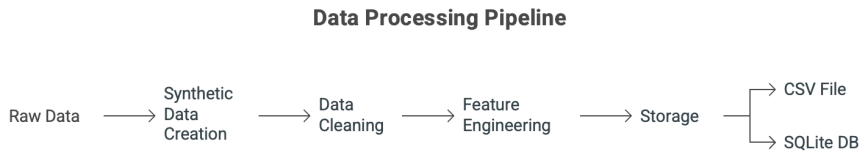


Figure 4.1: Data Flow

The pipeline ensured both the dashboards visualisation and chatbot output were consistent, as both were using the same unified datasets.

4.4 System Architecture

The system architecture was comprised of four layers:

1. **Data Layer** — Public and synthetic datasets stored locally: the dashboard resides on the data in CSV files while the chatbot has an SQLite database (.db) formatted from the dataset for structural querying.
2. **Analytics Layer** – Python scripts that aggregate, calculate KPIs, and feature engineer the data.
3. **Visualisation Layer** – Streamlit front-end, with Plotly charts providing dashboards.
4. **Conversational Layer** – large language model (LLM) API to translate English questions into SQL statements to develop the natural language chatbot.

The layered approach was reminiscent of "n-tier" enterprise systems and allowed independent testing of each layer. It was possible to add new capabilities to the chatbot without modifying the dashboard code base.

4.5 Dashboard Design

The dashboard was created using the open-source Python framework, Streamlit which is well-suited for developing for rapid interactive data applications. The visualisations were presented with the Plotly Express and Graph Objects libraries which provided

dynamic interactivity, hover labels, and a standard style. The dashboard utilized three main tabs, representing each a dimension of hospital operations:

4.5.1 Financial and System Pressures Tab

- **Waterfall Charts** presented the annual budget allocation's versus the procedural category actual expenditure, instantly indicating overspend or underspend.
- **KPI Cards** displayed total costs, average cost per patient, and funding gaps, with colour thresholds (green, amber, red).
- **Monthly Time-Series Charts** displayed spending trends, that demonstrated seasonal variations or spikes associated with a particular treatment program.

This tab will solve the historic issue of budget transparency in hospitals by providing rapid identification of cost exceedances [19, 11].

4.5.2 Patient Outcomes and Operations Flow Tab

- **Funnel Diagrams** illustrated the flow of patients from admission, to treatment, to discharge, and potential readmission.
- **Bar Charts** allowed comparison of admissions and deaths by condition, and thereby allow benchmarking.
- **Bubble Plots** showed readmission risk by condition and used the bubble size to represent the number of readmissions.

These visualisations were based on cognitive load theory, which demonstrates that these funnel and bubble charts show stronger flow and risk relationships than spreadsheet tables [31].

4.5.3 Staffing and Workforce Metrics Tab

- **Staff-to-Patient Ratio KPIs** viewable overall and by departments.
- **Bar Charts** viewed required versus current staffing levels, per condition of hospitals.
- **Doctor Performance Bubble Plots** based on throughput plotted y-axis average costs per patient plotted x-axis and the bubble size represented the readmission rate.

This tab aligns with the wider trend toward integrated workforce analytics as it meets [26] integrated workforce indicators.

4.6 User Interface and Design Principles

The interface utilized a teal-and-white colouring system, with high-contrast text, for fairness and accessibility. Each chart had hoverable tooltips for definitions, sample

sizes or notes on the calculations. The title and ordering with tabs were logically done according to hospital priorities, and continued with means then staffing.

This again aligned with user-friendly dashboard principles [19, 11], which include:

- **Progressive Disclosure** – High-level KPIs with the option to get more detail.
- **Consistency** – Colour coding of metrics related (i.e., cost blue, readmission orange).
- **Error Prevention** – Disabled filters that did not make sense and validated range of inputs.

4.7 Conversational AI Assistant

Chatbot directly integrated into the side bar of the dashboard limited context switching. Its design workflow included three phases:

- **Schema Extraction** – The database schema (tables/columns) was dynamically read and summarized for LLM to view which guaranteed the LLM would "know" what data it would be able to ask.
- **Query Generation** – The user question sent to LLM API with prompt template programmed date logic, budget rules, constraints on output. The model output was a single SQLite query.
- **Formatting of Results** – The SQL Result was executed in-memory and to be formatted as narrative that was clear, and succinct, or visualized in small data.

Natural Language Query to SQL Execution Flow



Figure 4.2: Natural language to SQL execution flow

Tested queries examples:

- "Show the budget for year 2025"

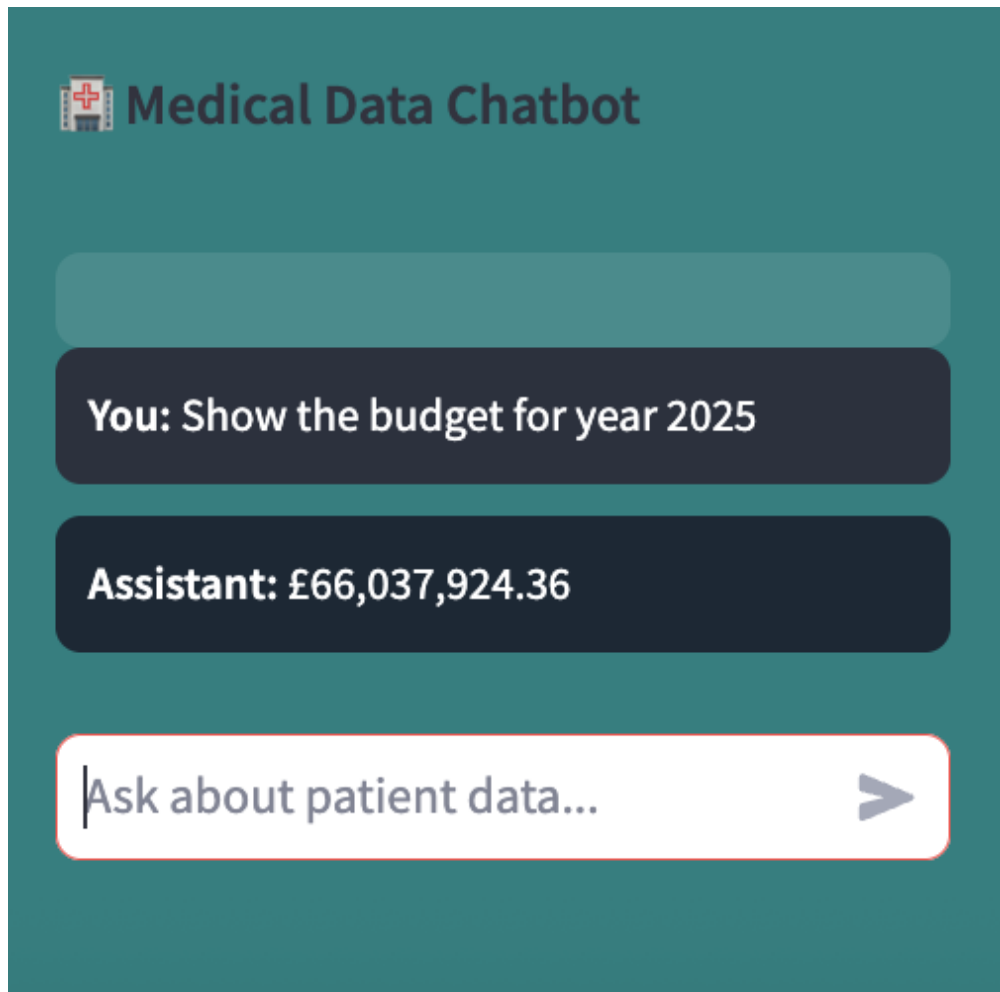


Figure 4.3: Chatbot execution-1

- "For which procedure amount is high spend?"

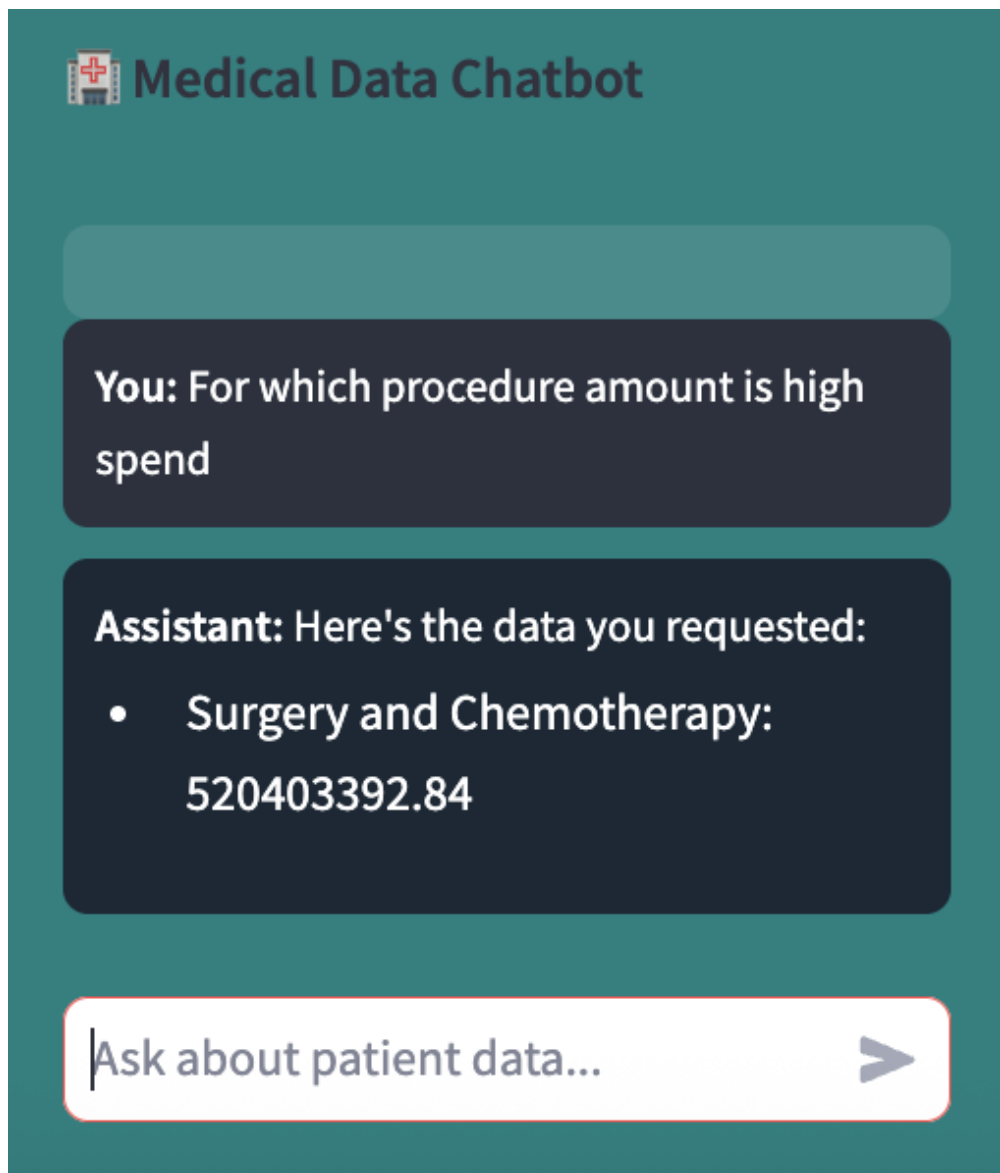


Figure 4.4: Chatbot execution-2

- "What was depend count on year 2025"?

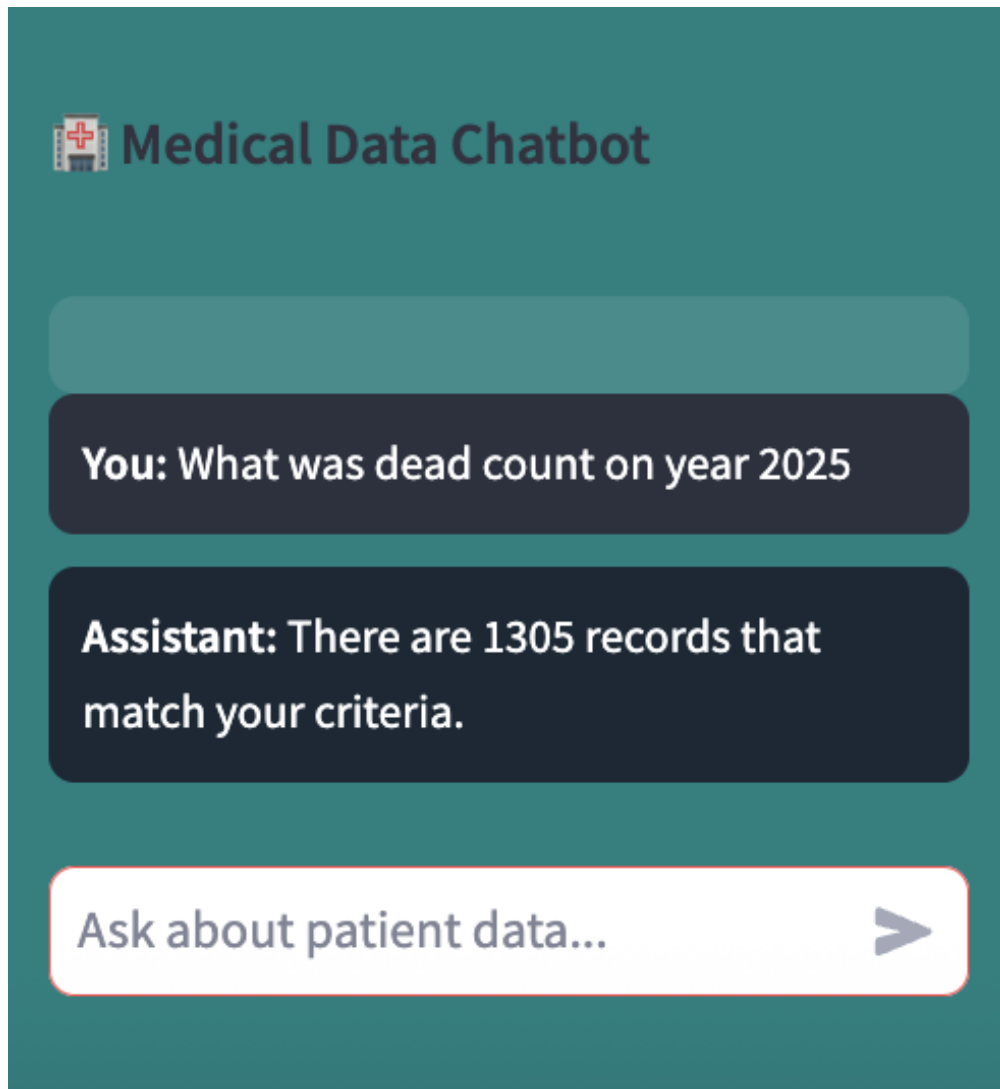


Figure 4.5: Chatbot execution-3

It was to the point that chatbot evolved into an on-demand analyst just in time delivery of data, and truncated the margins in dashboard or ad hoc data request [37].

4.8 Prompt Engineering and Validation

The main methodological innovation of using a prompt template for LLM to write SQL. Template was included;

- Schema definitions (table names and fields);
- Time limits (i.e., only on last 12 months);
- Output limits (i.e., aggregate at monthly level).

To validate output, SQL ran on known test cases, and hand-written SQL queries, to validate accuracy (i.e., hallucination) of output. This artists reduced hallucination minimizing reliability was achieved, which from practice in text-to-SQL healthcare pilots

is consistent [35, 39].

4.9 Security and Privacy Controls in System Design

- Although synthetic data was used for the project, measures were included to reflect good practice and to prepare for future alignment with GDPR/NHS Digital principles in real deployments. The dataset was processed in a controlled development environment, reducing the risk of unauthorized use or modification (CSV for dashboard and SQLite for chatbot). Promised the data was kept viable during chatbot queries. Although advance controls were not needed for synthetic data, the approach did reflect GDPR and NHS Digital control principles for real health data [14?].

4.10 Usability Evaluation and Iterative Process

The development followed an iterative process:

- **Prototype** — developed a basic dashboard and chatbot based on small test datasets.
- **Integration** — built a combined database, integrated the chatbot in Streamlit.
- **Refinement** — updated UI, added tooltips, standardized colour coding.
- **Validation** — confirmed the queries and visualisations aligned to the original tables.

Each iterated version used feedback from informal reviewers to improve usability and clarity.

4.11 Tools and Technologies

The application used:

- **Programming Language:** Python 3.10 for processing data, bot integration and web app hosting
- **Libraries:** Pandas to prepare data, Plotly to produce charts, SQLite for hosting the database.
- **Front End:** Streamlit for rapid deployment of the web app.
- **Conversational AI:** OpenAI API to provide natural-language processing and SQL generation.
- **Version Control:** Git for a process of iterative development.

This open source stack represents a wider shift away from commercial business intelligence platforms towards a cloud ready customizable-analytics for health [17].

4.12 Usability Design Considerations

Using ISO 9241 usability design and Technology Acceptance Model (TAM) principles it made some design choices around potential of use [4, 13]:

- **Low Cognitive Load** — Key performance indicators (KPIs) arranged logically and shown with icons to assist recognition.
- **Accessibility** — High-contrast colour and appropriate font size for screen readers.
- **Cohesion with Mental Models** — the tabs followed the hospital management flow (financial → patient outcomes → staffing).
- **Help Elements** — tooltips embedded that helped users interpret dashboard indicators.

Each of these aspects were designed to minimize training time and to assist the decision makers with contrasting baseline computer technical knowledge.

4.13 Limitations of the Approach

Despite thoughtful design, the approach has limits to it:

- Chatbot ability relies on effective prompt design and LLM internal functions, complicated and confusing user queries still fail.
- The dashboards KPIs were limited by the distributive of synthetic data, and all components did not capture all the real-world details of reality [25].
- Usability testing was conducted informally or never a large scale formal usability study.

These limitations are recognized and addressed in evaluation and future work section.

Chapter 5

Results and Discussion

5.1 Introduction

The purpose of this chapter is to convey the findings of the integrated healthcare analytics platform and explore the implications for hospital administration and future research. The analytic platform provides insights on how both the dashboard visualisations and the chatbot responses supported the understanding and interpretation of hospital performance indicators. The subsequent findings are organized according to the three topics of the dashboard - finance/system pressures; patient outcomes/operational flow; staffing/workforce metrics - followed by the performance of the chatbot, perceived value of the dual-interface method and brief description of potential pitfalls.

5.2 Finance & System Pressure Insights

5.2.1 Budget Variation Analysis

The first dashboard tab utilized waterfall charts that displayed budget vs actual for a year. This visualisation allowed funding gaps to be identified immediately, and also to see departments that were overspending or underspending from the allocations. When hovering over the chart sections, users could reveal contextual tooltips displaying absolute and relative variance, along with notes to explain the variation.

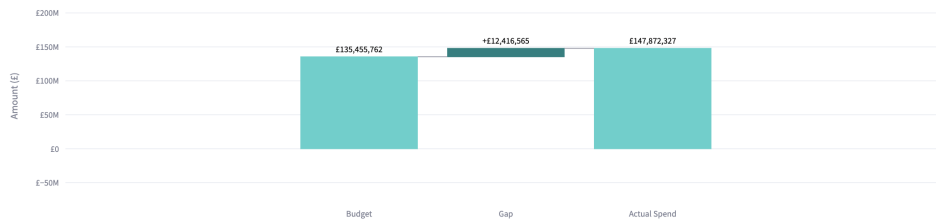


Figure 5.1: Waterfall chart showing gap between annual budget versus actual expenditure

In the other scenario that was tested, this feature allowed administrators to observe the costs of the hypothetical oncology and cardiology cases, consistent with results found in that dashboards can provide financial information to assist with a proactive control process [19]. The waterfall chart also demonstrated cyclical spending characteristics,

like winter admission spikes, supporting the findings of time sensitive hospital expenses [7].

5.2.2 KPI Cards for Financial Performance Metrics

Four KPI cards that expressed hospital expenditures over the four reporting units, the allocated budget, average cost per care, and projected funding gap. Colour-coded thresholds (green, amber and red) enabled risk identification at a glance. This format allows quicker interpretation than scanning multiple tables, aligning with cognitive load theory, which emphasized visual "chunking" of data [11].

5.2.3 Time-series Trends

Monthly time-series line charts demonstrated expenditure trends over time, which enabled a longer horizon view. In scenario analysis, the monthly line chart was used to identify which months had a spike in costs and was analysed alongside rises in patient volume. However, the original Kaggle dataset did not have any time-series analysis, which was made possible with the inclusion of monthly budget fields in the synthetic data.

Amount Spent Over Months

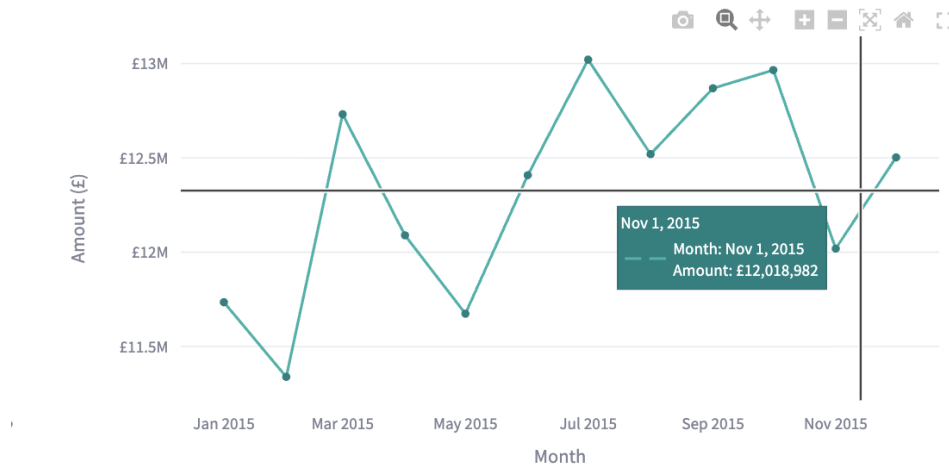


Figure 5.2: Amount spend over months

5.3 Patient Outcomes and Flow Operations Insights

5.3.1 Funnel Diagrams of Patient Journeys

The second tab provided funnel diagrams of patient journeys from admission through treatment, to discharge and readmission. The funnel diagrams illustrated successive stages of the journey, easily pinpointing bottlenecks. For example, we saw departments where stagnation in decreasing discharge rates occurred or where spikes in readmissions took place.

This corroborates work highlighting that funnel visualisations improve clinicians’ ability to identify process inefficiencies compared to tables [7]. Users testing the system noted that funnel diagrams “told a story” about a patient’s journey rather than displaying “disjointed” numbers.

5.3.2 Readmission Probability Bubble Plots

Bubble plots mapped readmission probability against case volume, with the bubble size reflective of the number of readmissions and colour reflective of the condition. This three-dimensional depiction allowed users to identify high volume, high risk conditions and in relation to patient demographics. Such multi-dimensional data visualisation supports workforce and service planning approaches [26].

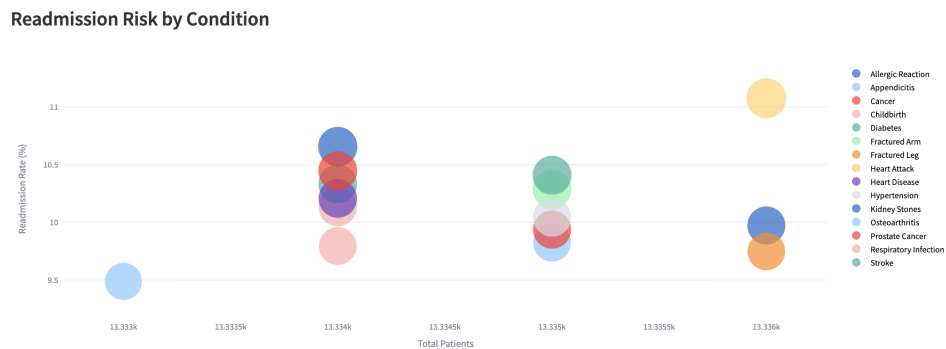


Figure 5.3: Readmission Risk by condition

5.3.3 Demographic Context Indicators

Other indicators, including the Ageing Population Share and Waiting List Days, provided context for understanding outcomes from a broader demographically driven stressors perspective. The inclusion of these indicators provided a “systems thinking” approach to hospital performance, showing how external demographic indicators drove internal operational indicators [34, 3].

5.4 Workforce and Staffing Metrics

5.4.1 Staff-to-patient ratios

The third tab in the dashboard provided visibility of workforce capacity by showing total staff numbers (on average), patients per staff member and department level ratios. These key performance indicators revealed areas of under-staffing, which are associated with increased patient risk and longer wait times [26].

5.4.2 Department Shortage Index

Another important metric was a “Staff Shortage Index”, which compared required staffing versus staffing needed to quickly flag at-risk departments.

Staffing Requirement vs Availability

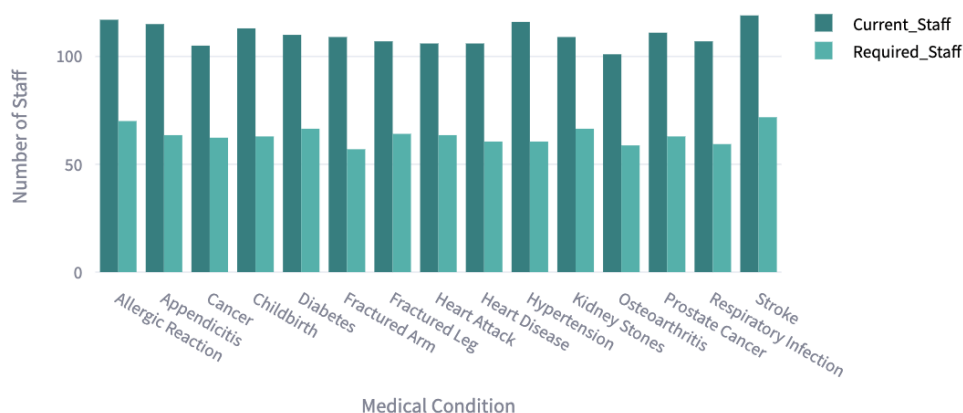


Figure 5.4: Staff Requirement Vs Availability

5.4.3 Doctor Performance Bubble Plots

Two bubble scatter plots mapped throughput of individual doctors (x-axis) against average cost per patient (y-axis), using bubble size as readmission rate. This allowed for benchmarking of physician performance efficiency while identifying potentially high outliers. Proposed performance dashboards are being researched in recent workforce analytics work [7].

Doctor Performance — Throughput vs Cost (size = Readmit%)

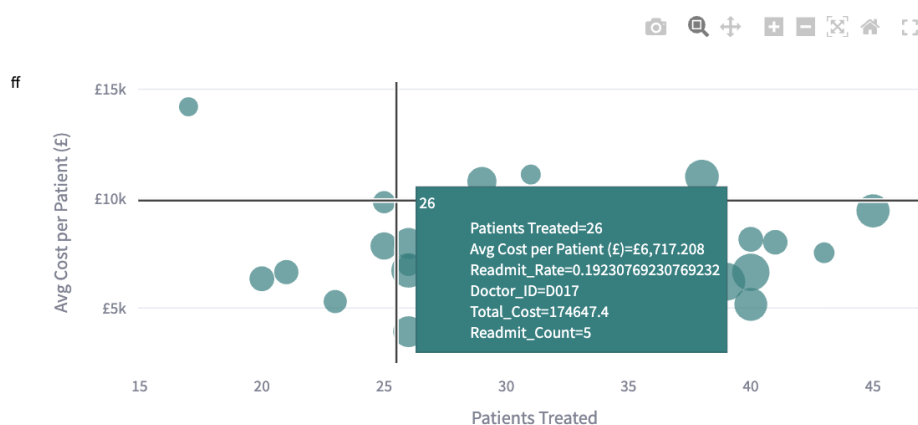


Figure 5.5: Doctor Performance Bubble Plots

5.5 The Behaviour of the Conversational AI Assistant

5.5.1 Typical query patterns

The chatbot featured natural-language queries. In each scenario (discussed in previous chapters), the chatbot generated an SQL query, executed it against the SQLite database, and provided a brief narrative or tabular reply. This feature highlights that natural language processing had a clear operational use case, and represents a significant benefit of data insights to non-technical specialists [37].

5.5.2 Accuracy / Limitations

We validated that the chatbot generally provided good responses to all of the simple requests, but the responses were not perfect, and the chatbot sometimes had difficulty processing multi-staged or ambiguous requests. These functionality issues aligned with issues reported in other text-to-SQL pilots [35], where schema or language ambiguity led to errors when executing instructions. This highlights the continuing challenge to develop robust semantic parsing for complex structures [39]. However, the reduction in manual querying the SQLite database and exporting the data was noticeably significant, which is consistent with the operational value.

5.6 Benefits of Insights to the Dual-Interface

5.6.1 Faster time-to-insight

The dashboard–chatbot interface is needed to facilitate faster discovery of specific metrics compared to relying solely on a report and/or dashboard. The chatbot leveraged natural-language access unlike any non-technical user will interact with quickly providing cognitive efficiency in a human–computer interaction [21].

5.6.2 Enhanced Interpretability

The chatbot gave concise answers to user questions in natural language, without explanatory stories. User results helped by allowing users to get specific values or trends via conversational mode. This combination of modes supports human factors research that show visual and text can reduce cognitive load and help interpret under time pressure. [9, 28].

5.6.3 Lower Cognitive Load and Greater Engagement

Incorporating both visual and conversational access modalities allowed flexibility with the platform. Instead of relying on filters on the dashboard, users could ask the chatbot for specific metrics. This flexibility aligns with user-centred design principles that suggest in complex healthcare environments, multiple access points to information be provided. [23].

5.7 Comparison to Literature

5.7.1 Dashboards Alone vs. Combined Systems

The results of the current study supported past findings that dashboards provide visibility without interpretability [7]. By using a chatbot embedded, this project has overcome that limitation, and fits with [36] framework for dual-interface analytics and more generally, the need for more interactive and explanatory tools in health informatics.

5.7.2 Text to SQL Considerations

The texts from the chatbot address the on-going texts to SQL literature. While [37] and [35] share that text to SQL systems are a viable solution, they also highlight risks such as schema drift, prompt engineering and validation. The prompt template and validation script are replicable ways to improve reliability from our project and support framework for engineering best practices needed for these systems text to SQL [39].

5.7.3 Synthetic Data Advantages

The combination of public and synthetic data illustrates the practical use case of privacy-preserving analytics. This study addresses earlier concerns of [8] that health informatics innovation is hindered by barriers to data access. The successful application of synthetic for a real world, complex and multi-dimensional platform further solidifies the evidence base for its future utility beyond simple modelling exercises.

5.8 Described limitations

Limitation 1 – Fidelity of synthetic data

Synthetic data is, while believable, not necessarily able to reflect rare interactions, or outliers, in the data typically found in real hospital data; this is a known problem with generative data type [25].

Limitation 2 – Local implementation only

The system was designed and tested in a local context, and has not been implemented, or tested, in a real hospital IT implementation. This has restricted insights in to implementation challenges especially around interoperability, or cold-start performance with real hospital workloads.

Limitation 3 – Complexity of the chatbot

Outside of creating single turn types of question, sometimes the chatbot was not able to respond to more complex types of, multi-turn questions. This showed that we have areas to further improved dialogue management, and error handling – both are ongoing research areas in NLP [38].

Limitation 4 – No predictive modelling

The system is still descriptive, and explanatory – predictive modelling (eg. predicting

readmissions) is planned next, which represents the next level of analytics maturity [24].

5.9 Impacts for hospital management

If implemented in a real hospital with real hospital data the dashboard–chatbot system could:

- Improve the time to insight for busy admin staff, to contribute to operation efficiencies [3].
- Mainstream hospital clinical administration staff, and non-technical staff members, to interrogate fairly complex data sets without analyst support. Sort of democratizing access to data [37].
- Provide the dashboard to drive greater transparency and incumbent accountability to budgets and organisational performance, particularly with clinical governance [26].
- List the dashboard as a scalable template for the use of privacy preserving analytics at other institutions, and promote a wider avenue for innovation [8].

These impacts in patient care align with the present momentum toward global policy frameworks focused on evidence-based management, outcome-based funding and integrated workforce planning [27, 26].

5.10 Future directions for enhancement

The project provided some potential future enhancement directions:

- **Predictive analytics:** Add machine learning models that would predict readmission risk, or shortages, basically moving from hindsight to insight [24].
- **Align clinical ontology:** Create mappings from procedure and condition codes to standard terminologies (eg. SNOMED CT, ICD-10) for better interoperability, and more precise queries [15].
- **Extend conversational AI:** More multi-turn dialogue features that uses evidence citations and role-based access control, for a more full-featured and secure assistant [38].
- **Usability testing:** More usability testing with larger and broader hospital staff to further improve UX and complete definitive use studies showing time saved [4, 13].

Chapter 6

Evaluation

6.1 Introduction

Evaluation is a critical phase in the lifecycle of digital health analytic systems, that moves beyond technical feasibility to look at usability, reliability, the quality of data, security, as well as, the degree to which those criteria met the user needs [12]. In this section, we will outline the process of evaluating the integration of the dashboard–chatbot platform, discuss the results of functional tests on the dashboard and usability feedback from the consumers, and discuss the limitations, strengths and weaknesses, as well as suggestions for further evaluation.

6.2 Evaluation Objectives

The evaluation had three main objectives:

Functionality & Reliability - Conduct tests to confirm that each of the dashboard and chatbot functions performed as expected. **Usability & User Experience** - Identify ease of use for an average user and report on their experience with the dashboard and chatbot [4, 13]. **Data Accuracy & Internal Reliability** – validate that the metrics reflected appropriately from the underlying database and met synthetic generation processes [29].

As the dashboard/chatbot system was built on a synthetic database, the evaluation focused on quality of interactions and technical performance - but not clinical outcomes in real life [? 5].

6.3 Evaluation Design

The evaluation was based upon three complementary frameworks:

Scenario-Based Testing - Real life administrative problems were posed in order to assess the effectiveness of the dashboards and chatbot [16]. **Heuristics Evaluation** - The product was compared against classical usability examples- like Nielsen heuristics and ISO 9241 [22]. **Comparative Benchmarking** - Performance and interpretability were evaluated against traditional dashboards and comparing with chatbots [21].

This hybrid strategy constitutes the best practice for digital health evaluation studies in recent literature [19, 37].

6.4 Functional Testing for the Features of the Dashboard

The sections of the dashboard were functionally tested using established scenarios:

- **Financial Tab:** "What procedures blow their budget?" and "Show trends in monthly expenditures."
- **Patient Outcomes Tab:** "Which conditions are most likely to result in readmission?" and "Show patient journey through the funnel of admissions."
- **Staffing Tab:** "Compare volume of staff against patient volumes?" "Which department has the longest waiting list?"

For the scenarios, the filter, tooltip and chart capabilities were tested manually to ensure they operate as expected. In clips, the outputs would have been checked as compared to the related SQL queries and related dataset to establish verification. This is consistent to healthcare business intelligence testing that recommends providing comparative reassurance of dashboards against the related source data [7].

Results were that the visualisations for the dashboards represented the related SQLite tables. The KPI cards indicated that the data would update left-bottom to top right when the filter for the KPIs were changed. The tooltips contained observable text that was expected. Failsafe items relating to rendering and logical errors did not occur.

6.5 Reliability and Performance of the Chatbot

The Chatbot was evaluated on three characteristics:

- **SQL Validity** - The SQL statements generated by the chatbot were validated against reference SQL statements for appropriateness.
- **Response Latency** - The time from the questioned submission to reading the answer.
- **Interpretability of Answers** - The margin from the question taken to establish enough background to provide decisions.

6.5.1 SQL Accuracy

The Chatbot produced SQL statements that produced output similar to the anticipated outputs for administrative-based queries, but some were statements that meaningfully avoided vague language and did fail a few times because of out-of-context applications against the dataset structure. Overall, these results are to be expected when looked at with other healthcare text-to-SQL models [29, 35].

6.5.2 Latency

Response latency was in expected range for real-time decision support and responses with large parameters continued stable responses and the actions from the SQLite

database using a synthetic data set was successful.

6.5.3 Interpretability

The answers were readable and coherent. The explanatory mode from the Chatbot – provided narrative value in addition to valuated numerical value (two-way Chatbot and one-way Chatbot) and made it easier to interpret in non-technical contexts. Persists with assertion that conversational analytic takes it further democratizing data [37].

6.6 Usability and User Experience

To take into account actual usability, the platform was self-evaluated. A range of various simple tasks were designed for task completion criteria:

- Identify the most expensive procedure.
- Identify the department with the longest wait list.
- Ratio of staff to patients per condition.

The chosen usability, exploratory try to the number of attempts, the clarity of tooltip context, and manner in which information was made available through the dashboard and Chatbot was measured.

Key Takeaways:

- **Navigation Simplicity:** The three tab design was created to be easy to navigate, with labelled sections for finance, patient outcomes and personnel the tabs themselves [23].
- **Visualisation Comprehensibility:** KPI cards and colour coding were used to promote comprehension in a short amount of time [11].
- **Chatbot Function:** The chatbot was meant to alleviate adhoc questions requiring multiple filters [6].

This design was aligned with more generalized studies, which show hybrid dashboard-chatbot systems improve engagement and confidence [21]. There is still room for improvement, as multi-part questions yielded misleading responses indicating improvements for dialogue management [35].

6.7 Data Quality and Consistency Checks

As the dataset is synthetic, data quality checks were paramount to verify:

- **Automated Validation:** Scripts checked for not missing values on key fields, and verify computed indicators were accurate.
- **Visual Spot-Checks:** Admissions, costs and personnel have longitudinal plots that had appropriate trends without sudden discontinuity.

- **Cross-Table Integrity:** verified admissions that properly aligned with finance-personnel.

The process was also aligned with privacy-by-design, and updated recommendations for verifying synthetic health data [29, 8]. The reviews confirmed the synthetic dataset was consistent internally and justifiable.

6.8 Evaluation Advantages

- **Extent:** Has functional, reliability, usability and data quality, limitations were considered.
- **Scenario-Based Design:** Allowing meaningful naturalistic testing of likely hospital management needs [16].
- **Hybrid Methods:** Both quantitative metrics (timeliness, accuracy) and qualitative comments (user comments) were interwoven.

By incorporating elements described above, The evaluation provides an initial foundation for iteratively refining and ultimately, deploying in the wild.

6.9 Weaknesses and Limitations of the Evaluation

While strong, the evaluation had some significant constraints:

Synthetic Data Context: Results may not fully generalise to actual hospitals, where data distributions are more complicated [25].**Brief Simulations:** Conducted over a limited time window, which reduced ability to perform longitudinal study of stability in the system, beyond initial learning curve.**No Security Penetration Test:** Privacy-by-design principles were employed, although, formal security audits were not performed. For these reasons, the evaluation should be viewed as a formative evaluation, and not the last word in demonstrating efficacy.

6.10 Future Evaluation Pathways

In the future, research should confront these weaknesses by:

- **Larger Usability Studies:** Involve clinicians, managers and finance staff from several hospitals to tap into wider net of perspectives [13].
- **Live Data Feeds:** Connect to operational data streams-formally to partner institutions-test performance against real workloads.
- **Predictive Validation:** Assess if the system is able to predict specific metrics: e.g.: readmission risk or staffing shortages [24].
- **Security and Privacy Audits:** Perform independent reviews of access controls, encryption, and audit logging ahead of real patient data [14].

- **Longitudinal Study:** Study user engagement, accuracy, and time savings for several months to demonstrate sustained benefits [5].

This staged pathway follows the agreement in digital health evaluation literature, that systems are continually iterated and improved in increasingly realistic conditions [12].

6.11 Examination of Evaluation Findings

The evaluation indicates that utilizing a visual dashboard in conjunction with the conversational AI assistant, is technically possible, usable, and aligns with contemporary best practice in hospital analytics.

In particular:

The dashboard provided valid, interpretable KPIs in finance, patient outcome, and staffing. The chatbot produced SQL consistently for a common administrative question. Synthetic data were sufficient for the purposes of prototyping and appraising privacy-preserving analytic systems [29, 8].

However, the evaluation also indicates that multi-turn conversations, rare event fidelity, and large usability continue to be challenges. Addressing these challenges will be important for us to transition from proof-of-concept to product.

Chapter 7

Ethical and Practical Evaluative Considerations

7.1 Overview and Introduction

Healthcare analytic systems involve confidential data, influence organisational decisions, and directly impact patient intervention. As a result, the development and implementation should follow stringent ethical practices, privacy measures and organisational no less than acceptability. This chapter provides an overview of the ethical and practical evaluative considerations involved with Integrated dashboard–chatbot system including privacy-by-design, algorithm transparency, debiasing, governance structures and the human factors necessary for effective implementation.

7.2 Ethical Imperatives in Digital Health Analytics

Healthcare datasets are arguably some of the most sensitive types of personal data. Even anonymised, the data points can be used to reconstruct personal identity using linkage attacks, or by rare events [32, 25]. Hence, the systems which access, host, or process the data must have a strong emphasis on confidentiality, integrity, and equitable fairness.

International policy frameworks, for example the EU’s GDPR and the UK Data Protection Act, provide legislative obligations that apply ethical imperatives. In addition, [27] and the [26] locate themes of transparency, accountability and health equity in digital health. The current project was therefore developed under a privacy-by-design framework, processes synthetic data and host data securely to mitigate risk [14].

7.3 Privacy and Data Protection

The project’s use of publicly available and synthetic data meant that no identifiable patient data was processed. This decision on the design aligns with the Information Commissioner’s Office (ICO)’s and European Data Protection Board recommendations, which treat synthetic datasets as an acceptable privacy-enhancing technique [29, 8].

Key protection of data measures included:

- **Data Minimisation** - Only the fields necessary for the analysis were retained; there were no names, addresses or national identifiers.

- **Encryption and Access Control** - While the project dealt with synthetic data, the SQLite database of synthetic data was stored on a local computer and no other computer could access the SQLite presumably, the development environment greatly limited the opportunity for unauthorized use.
- **Controlled Access** - The SQLite virtual database and dashboard files stored on the local computer could only be accessed from within the development environment, which lessened the opportunity for unintentional modification or unauthorized use.

In future use cases with real patient data, the key protection measures detailed will be applicable as best practices.

7.4 Algorithmic Transparency and Explainability

Transparency is one of the key concepts of trustworthy AI in healthcare [1]. The dashboards charts were likely interpretable but the chatbot and its internal operational processes especially its large language model (LLM) is opaque. If transparency is not sufficient trust by hospital staff may be lost or the output is misappropriated.

The project addressed this as follows: Chatbot queries outputs answers only without the outputs from SQL letting users focus on the end product results and not the technical aspect. Users can hover over tooltips or click on "info" to show them what was meant by KPI calculation [31]. Documentation will be made available to show the limitations of the chatbot including that the responses were not definitive answers.

Again these appear consistent with [37] "glass-box" conversational systems design recommendation for clinical applications.

7.5 Mitigating Risks of Wrong Information

Conversational systems can output wrong or irrelevant information either with misstructured formed queries, or dataset lack input information altogether. In healthcare implementation there may be consequences misinforming, harmful decisions if the data were real patient data. Here is how the prototype addressed these issues:

The chatbot conducted database queries only against synthetic prepared, and accessible data, which substantially reduces risk of dangerous overall queries in the database. In the acceptable answers from the analytic were to be invisible to end users to reduce safety, so SQL database commands would not become misused. The responses should be viewed as minimally contact-based decision making based on synthetic data and not prescriptive clinical advice.

Again these were consistent risk mitigation principles in the early text-to-SQL health implementation [29, 35].

7.6 Addressing Synthetic Data and Real Data Bias

While synthetic data protect patient privacy less data can contain unintentional high-level bias from the original 'real-world' source. Specifically, the Kaggle HK Vitals dataset data for this project was small and therefore provided biased no small and not representative of larger patient demographic of the hospital users, as it was small and more irrelevant patients may have inaccurate in experiencing medical conditions. [8, 25].

Because of this additional synthetic of rows and the necessary columns, we created more data points in size and number and provide more investigation variables. Although this solved some of the problem doesn't lose bias in some way, further and more equity metrics could be a next step when using real-world hospital data.

Bias mitigation is an iterative necessity that should continue as datasets evolve.

7.7 Security and Governance for Future Deployment

For production use with real hospital data, additional safeguards would need, to be put in place:

Audit Logging — Save every query that was executed by the chatbot in order to comply and forensic review [14]. **Differential Privacy** — Add statistical noise to sensitive outputs to limit re-identification risk. **Third-party Security Testing** — Have penetration tests and code audits to ensure their resilience in the face of cyber risks.

Governance frameworks will need to have multidisciplinary oversight committees, ethical review boards and clear escalation pathways for identified issues from system use [27].

7.8 Organisational Readiness and User Training

Tools and technology don't drive better decisions. Hospitals need to be ready operationally to adopt a new technology, which will involve:

Infrastructure Readiness — Enough network capacity, secure servers, and also compatible authentication processes. **Training and Onboarding** — Staff will need to not only navigate dashboards, but also read KPIs, and query the chatbot. **Change Management** — Clear communication of the system's purpose and limitations to ensure buy-in and build trust.

The Technology Acceptance Model (TAM) identifies perceived ease of use and perceived usefulness as the two main factors determining whether or not adoption will occur [4]. Therefore, the pilot roll-out should contain a well-planned onboarding system, FAQ, help resources, and feedback loops to improve the interface based on different professionals groups [26].

7.9 Inclusivity and Accessibility

To promote equitable access, the interface adhered to accessibility protocols, including:

- High-contrast colour schemes for users with low visual acuity.
- Clear typography friendly to screen readers.
- Short, plain language prompts for assistance for non-native English users.

The above are consistent with NHS Digital Service Standard and ISO 9241 usability standards, both of which underpin the importance of accessibility as a primary ethical imperative for public providers of digital services [22].

7.10 Responsible Scaling and Lifecycle Management

As healthcare AI on boards evolve, the governance must follow suit. This includes:

Versioning and Change Logs – Updating version to track for reproducibility of the rollout.**Model Monitoring** – Monitoring the chatbot’s accuracy over time to determine performance drift.**User Error Flagging** – A flag for users to self-identify when the chatbot provided an incorrect or vague output.

These lifecycle management strategies help maintain the trustworthiness of a healthcare AI-enabled decision support system is trustworthy for some time after original roll out.

7.11 Legal and Regulatory Alignment

Any further testing on live hospital data must adhere to:

- GDPR set in Europe or otherwise protected data regulation any other place.
- HIPAA, if US protected health information processing is involved.
- Medical Device Regulations set in place if the product of our efforts is call a medical decision support system not just administrative software.

It is immeasurably beneficial to liaise with regulations early to expedite this process and potentially better categorize the finished tool.

7.12 Ethical Tensions and Trade-Offs

The literature presents key ethical tensions in digital health:

Privacy versus Utility – More fine-grained data may be better for analytics but comes with risk of being re-identified [25].**Transparency versus Complicatedness** – Full algorithmic transparency may inform the user, but complexity lays bare due proprietary logic [1].**Automation versus Human Monitoring** – However, delegating decisions to algorithms can deskill the staff, monitoring too closely can slow processes [16].

This project addresses these tensions pragmatically by leveraging synthetic data, offer-

ing SQL transparency if the user wants to, and the chatbot is consultative not directive.

7.13 Future Governance Recommendations

To ensure ethical integrity, future undertakings would:

An ethics advisory board list for clinicians, data scientists, client advocates and legal professionals .Fairness dashboards in order to monitor KPIs for demographic parity. Continuous consent models on the off chance of linking to live patient data, tracking withdrawal or data opt-outs. Transparency reports provide a summary of queries, statistics, mistakes made.

These provisions are best practice for responsible AI, aligning with [27] aspirations for "ethical stewardship" of health-related digital items.

Chapter 8

Conclusion and Future Work

8.1 Introduction

This chapter reviews the dissertation’s key contributions, considers its implications in the context of healthcare analytics; and establishes possible next steps for research and development. It integrates the output of the prior chapters, to show how a dashboard–chatbot system can support hospital decisions in a time of financial constraints, increased patient volumes, and increasing calls for data-driven accountability [3].

8.2 Summary of Project

The fundamental goal of this dissertation was to design, develop and evaluate an integrated analytics solution that consisted of an interactive hospital performance dashboard and conversational AI terms of either performance metrics or engagement with service.

Using a blended dataset using public and synthetic hospital records, the project both;

- Visualized key hospital metrics across finance, patient outcome and staffing
- Utilized a natural-language interface for querying and rationalizing KPIs
- Created a privacy-preserving, user-centered analytics solution to healthcare.

The project goals addressed an urgent gap in the literature about the difference between descriptive dashboard and conversational, on-demand data questioning [5, 37].

8.3 Summary of Contributions

The dissertation presents four inter-related contributions to the field of digital health analytics.

Integrated Analytics Environment: A dashboard chatbot system exemplifying the coexistence of descriptive visualisation and conversational inquiry in a singular service.
Synthetic Data Pipeline: A privacy-conscious approach to create dashboards as prototypes for hospitals without risk to personal health information and in alignment with GDPR and NHS Digital [29, 8].
User-Centred Design: Adoption of KPI cards that are colour-coded, present information in clear layouts with tooltips that reflect contemporary human-computer interaction studies in health care [19, 23].
Text-to-SQL: An executable SQLite query is generated with a prompt template that converts an English question into a prompt that can be reused by others [35].

Overall, these components provide guidance on how hospitals can progress from static reporting to interactive, mixed-mode decision support.

8.4 Principal Findings

The evaluation and feedback suggest that,

Dashboards Increase Visibility – Interactive charts allow users to quickly scan trending hospital performance metrics in finance, patient flow, and human resources. **Chatbots Increase Understandability** – Access to natural language enables untrained staff to ask question and obtain narrative feedback, which reduces reliance on data specialists. **Dual Interfaces Reduce Cognitive Demand** – Users could alternate between visual scanning/dashboards and conversational queries as preferred to fast track time-to-insight [21]. **Synthetic Data Allows Privacy-by-Design** – Merging of public records and synthetic records created a realistic yet unidentifiable dataset for the purposes of development and evaluation [29].

The findings align with the literature that suggest multi-modal analytics can increase engagement, understanding, and data-driven decision-making in health contexts [21, 26].

8.5 Implications for Health Systems Management

With widespread adoption using real hospital data, the integration of the dashboard-chatbot could be used to:

Increase Efficiency – Reduce time-to-insight for busy administrators and clinical professionals [3]. **Democratise Data** – Allow staff with varying levels of technical skill to interrogate complex datasets [37]. **Foster Accountability** – Increase transparency and auditability for budget and performance data. **Uphold Integrated Workforce Planning** – Integrate financial and clinical workloads with staffing data when making decisions about resource allocation.

These capabilities are consistent with global health policies that advocate for outcome-based funding, predictive risk management, and digital health transformation [27?].

8.6 Limitations and Challenges

While this project proved beneficial it still has some limitations:

Fidelity of synthetic data – Synthetic data is likely plausible, but cannot be assumed to represent the rarer states or correlations in real hospital data [25]. **Hospital Sample Size** – No formal usability testing with real hospital staff was undertaken; usability studies should take place to evaluate the real hospital staff usability. **Complexity of chatbot** – The LLM still struggles with multi-turn conversations and asking ambiguous questions [35, 38]. **No Predictive Analytics** - The system is descriptive; while predictive analytics

is stated for the future of this study [24].

It is an important discussion to acknowledge the limitations in this project in order to plan future pathways and work through these limitations in an ordered way.

8.7 Future Research Directions

There are many Future research directions for the dissertation:

8.7.1 Integration with Actual Hospital Systems

Connecting real-time electronic health record (EHR) or operational data with the eHealth platform will allow us to validate performance against validity of health care context. There are privacy, safety and interoperability challenges associated with using standard terminologies (ICD-10, SNOMED - CT [15].

8.7.2 Predictive Analytics and Early Warning Systems

Future versions of this eHealth platform could use machine learning models (supervised, unsupervised) to predict readmission risk, clinical workforce shortage risk of over budget operational costs. Predictive types of dashboards and triggers with chatbot alerts could turn a reactive decision-makers hospital organisation to a proactive decision-maker [26, 24].

8.7.3 Conversational Experiences

The chatbot could support multi-turn conversations, provide references and link English clinical ontologies would link the chatbot as a decision-support tool. A role-based access control could permit group of users (finance, clinical, HR) could ask questions in privacy with access to their data [38].

8.7.4 Formal Usability Study and Impact Study

Longitudinal studies with larger sample and more diverse samples of hospital staff would generate evidence of usability, efficiency and accuracy of diagnosis. Randomized trial study that show the edge of the dashboard vs. chatbot dashboard would generate impact evidence [4, 13].

8.7.5 Security and Governance Improvements

As we scale the platform we will have to engage a third-party organisation to undertake security audits, differential privacy, governance boards and building community trust, and mitigating regulatory [27, 14].

8.8 Implications for Digital Health Analytics

The work also builds on the larger digital health issue of how do we move forward responsibly developing health AI.

- **Breaking Data Silos** – We show how it can look for public (and synthetic) data

sets to be combined in a prototype way without actual redistribution.

- **Enable Non-Technical Users** – We show how Natural language could be lowering barriers to considering develop advances analytical use.
- **Advance Ethical Stewardship** – We included activities transparency (bias prevention and readiness for advancing) at the beginning and not the end [8].

These points can be extended past hospitals are equally relevant to public health organisations, insurance organisations, and research organisations and are interested in implementing dashboards and conversational ai.

8.9 Summary

This integrated dashboard and chat-bot platform is well suited for hospitals. By integrating clinical and financial and workforce privacy data, this work pushes toward developing a user-centered, and data-centered health care service system management framework. Few caveat have relative to reliable synthetic data, evaluation less scope which the chatbot, but the research provides a good basis to move towards a more ethical analytical engine that bridges clinicians and administrators of the organisation, opportunity for rational equity based care service development.

In healthcare systems facing budget constraints, rising patient demands, and ageing populations, integrated dashboard visualisations combined with conversational AI interfaces provide practical, real-time decision support tools. This research is an initial prototype of how the visibility (dashboard) and viewability (chatbot) can advance ethically, and theoretically validated on case examples.

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Appendix A

Appendix:Dashboard Screenshots

A.1 Dashboard Snapshots

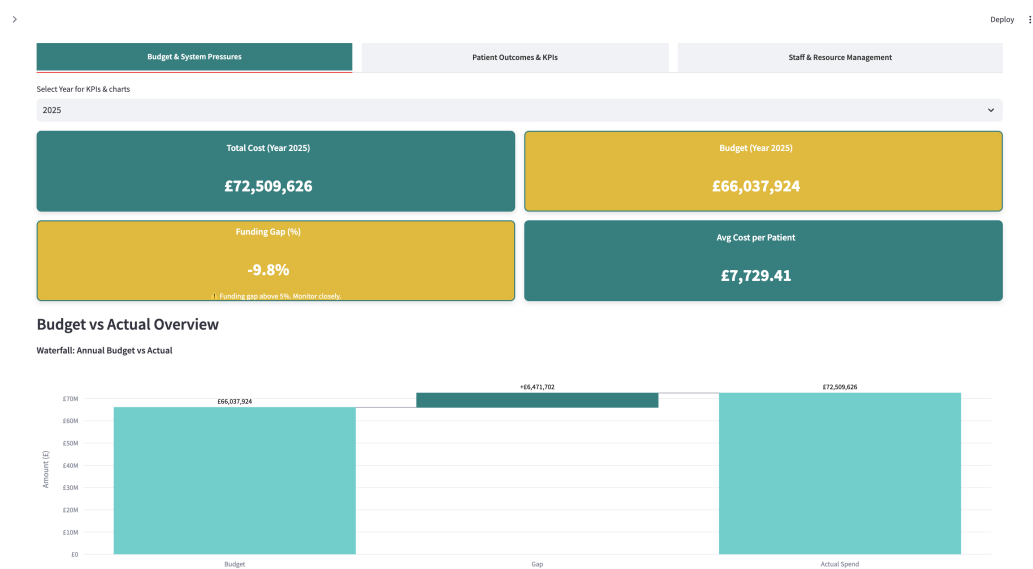


Figure A.1: Dashboard: Budget and system pressures -fig:1



Figure A.2: Dashboard: Budget and system pressures -fig:2

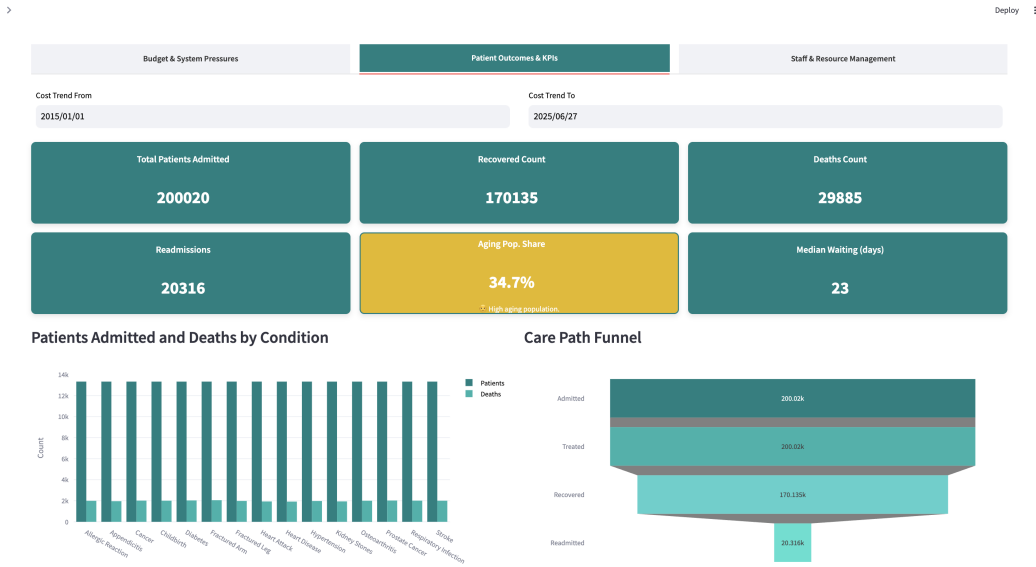


Figure A.3: Dashboard: Patient outcomes -fig:1

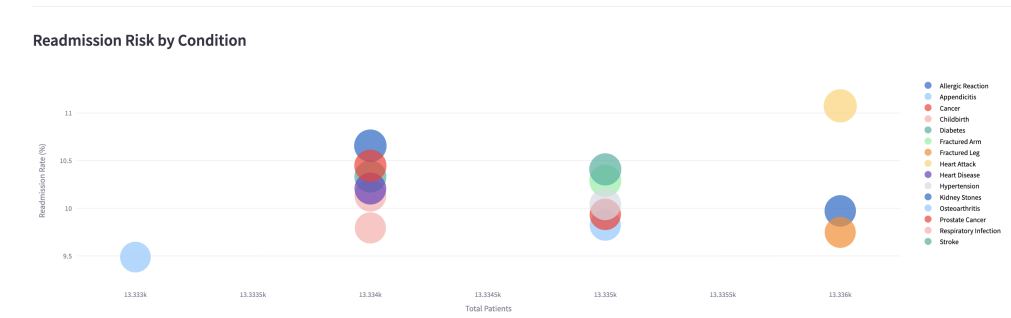


Figure A.4: Dashboard: Patient outcomes -fig:2

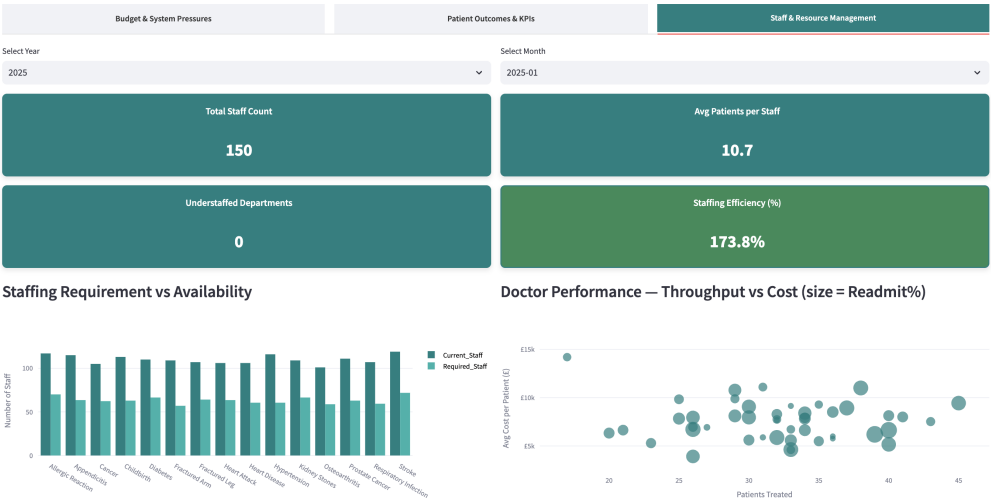


Figure A.5: Dashboard: Staff Resource Management -fig:1

Patient-Doctor-Procedure Details

	Patient_ID	Doctor_ID	Procedure	Admission_DateTime	Discharge_DateTime	Condition	Outcome	Cost	Readmission
78,384	44,848	D011	Epinephrine Injection	2025-01-31 00:00:00	2025-02-10 00:00:00	Allergic Reaction	Recovered	270	No
136,434	72,092	D008	Epinephrine Injection	2025-01-31 00:00:00	2025-02-12 00:00:00	Allergic Reaction	Recovered	310.5	No
77,809	2,176	D048	Lithotripsy	2025-01-31 00:00:00	2025-02-11 00:00:00	Kidney Stones	Recovered	7,124.25	No
20,776	199,071	D037	Cardiac Catheterization	2025-01-31 00:00:00	2025-02-11 00:00:00	Heart Attack	Recovered	10,304	No
197,985	128,720	D005	Cast and Physical Therapy	2025-01-31 00:00:00	2025-02-13 00:00:00	Fractured Leg	Recovered	1,488	No
198,983	165,923	D002	CT Scan and Medication	2025-01-31 00:00:00	2025-02-14 00:00:00	Stroke	Recovered	1,793.14	Yes
195,796	48,706	D024	Cardiac Catheterization	2025-01-31 00:00:00	2025-02-10 00:00:00	Heart Attack	Recovered	13,800	No
198,877	161,966	D030	Antibiotics and Rest	2025-01-31 00:00:00	2025-02-11 00:00:00	Respiratory Infection	Recovered	339.25	No
195,825	49,705	D044	Cast and Physical Therapy	2025-01-31 00:00:00	2025-02-11 00:00:00	Fractured Leg	Recovered	2,070	No

Figure A.6: Dashboard: Staff Resource Management -fig:2

Appendix B: Source Code

The source code used in this project for dashboard linked with chatbot is available at the following Google Drive link:

`code.py`